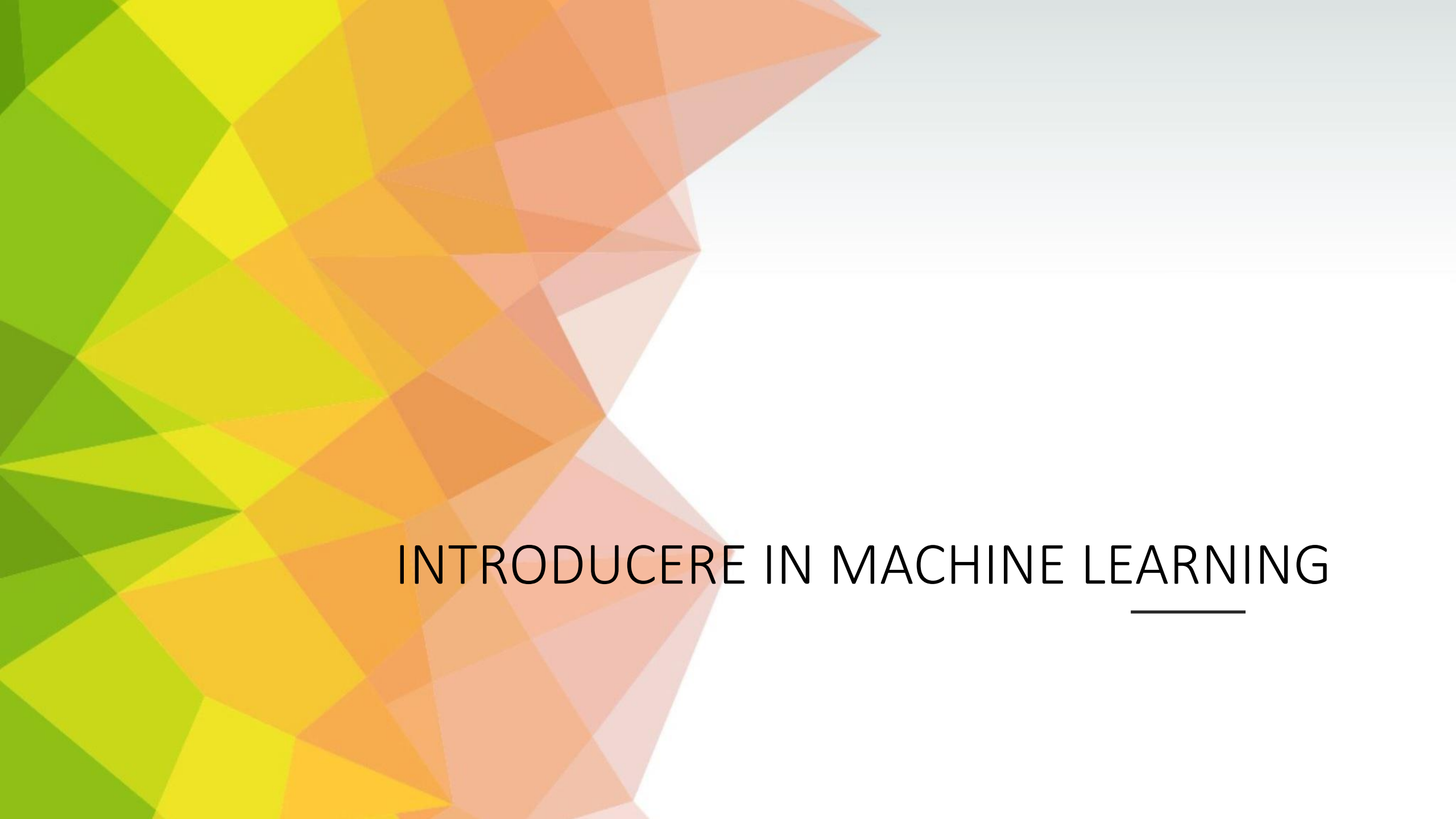




INTRODUCERE IN MACHINE LEARNING

Conceptul de ML (invatare automata)

ML reprezinta procesul de antrenare prin modelare a unui software pentru a face predictii asupra unor date.

In rezolvarea unei probleme ML seamana mai mult cu modul in care invatam noi oamenii.

Invatare prin exemple, invatare bazata pe experienta acumulata.

Calculatorul invata prin exemple, fara a fi explicit programat.

Abordare traditionala vs. invatarea automata

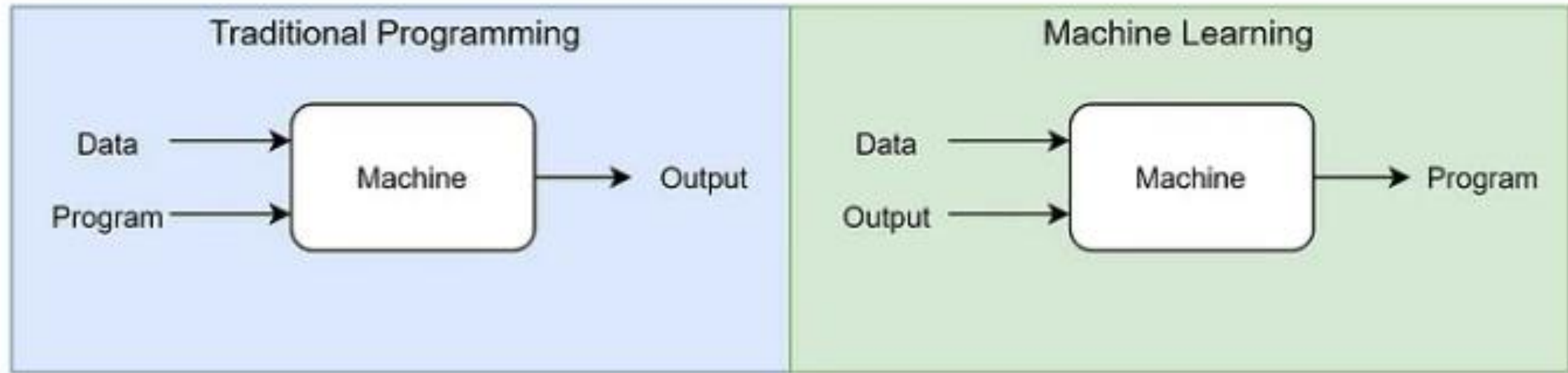


Figure 1: Traditional Programming vs Machine Learning

“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .”

Tom Mitchell

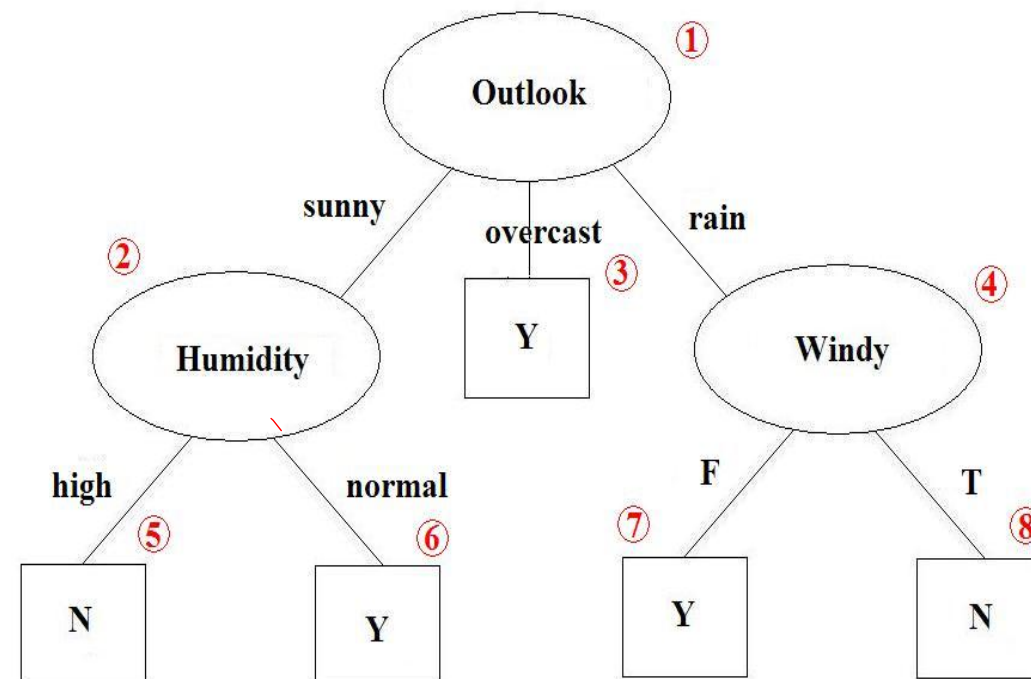
Exemplu

atribute intrare

atribut ținută
Output.

Outlook	Temp	Humidity	Windy	Play
sunny	hot	high	F	N
sunny	hot	high	T	N
overcast	hot	high	F	Y
rain	mild	high	F	Y
rain	cool	normal	F	Y
rain	cool	normal	T	N
overcast	cool	normal	T	Y
sunny	mild	high	F	N
sunny	cool	normal	F	Y
rain	mild	normal	F	Y
sunny	mild	normal	T	Y
overcast	mild	high	T	Y
overcast	hot	normal	F	Y
rain	mild	high	T	N

overcast mild high F. (?)



MODELARE

- Algoritmii ML construiesc un model matematic bazat pe o colectie de date denumita multime de antrenare (multime de training) pentru a face predicții, fără a fi programați în mod explicit să facă acest lucru.
- Scopul principal al ML este de a permite computerelor să învețe autonom fără intervenție sau asistență umană și să ajusteze acțiunile în consecință.

- **Data security:** Machine learning models can identify data security vulnerabilities before they can turn into breaches. By looking at past experiences, machine learning models can predict future high-risk activities so risk can be proactively mitigated.
- **Finance:** Banks, trading brokerages and fintech firms use machine learning algorithms to automate trading and to provide financial advisory services to investors. Bank of America is using a chatbot, Erica, to automate customer support.
- **Healthcare:** ML is used to analyze massive healthcare data sets to accelerate discovery of treatments and cures, improve patient outcomes, and automate routine processes to prevent human error. For example, IBM's Watson uses [data mining to provide physicians data](#) they can use to personalize patient treatment.
- **Fraud detection:** AI is being used in the financial and banking sector to autonomously analyze large numbers of transactions to uncover fraudulent activity in real time. Technology services firm Capgemini claims that fraud detection systems using machine learning and analytics [minimize fraud investigation time by 70% and improve detection accuracy by 90%](#).
- **Retail:** AI researchers and developers are using ML algorithms to develop AI recommendation engines that offer relevant product suggestions based on buyers' past choices, as well as historical, geographic and demographic data.

Aplicatii

[Sursa:](#)

[What Is the Definition of Machine Learning? | Expert.ai | expert.ai](#)

Exemple

Common business use cases 



Spam versus regular email

Recommended items



Recommendations



Fraud

© 2020 Amazon Web Services, Inc. or its affiliates. All rights reserved.

12

Computer vision 



PERSON 99.3%

OUTDOORS 83.1%

CREST 83.0%

MOUNTAIN BIKE 99.1%

ROCK 82.8%

© 2020 Amazon Web Services, Inc. or its affiliates. All rights reserved.

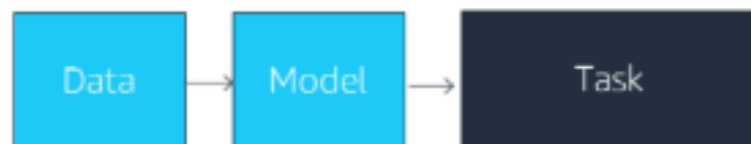
13

When to use machine learning?

Classical programming approach



Machine learning approach



Use machine learning when you have:

- ✓ Large datasets, large number of variables
- ✓ Lack of clear procedures to obtain the solution
- ✓ Existing machine learning expertise
- ✓ Infrastructure already in place to support ML
- ✓ Management support for ML

Tipuri de invatare

Supervizata (supervised):

- Algoritmul invata sa faca predictii plecand de la o multime de training care contine date deja clasificate (etichetate).
- Exista un atribut tinta.
- Daca a fost antrenat suficient va putea sa clasifice date noi, neclasificate, adica sa faca predictii asupra unor date noi, pe care nu s-a antrenat.

Nesupervizata (unsupervised):

- Proces independent de invatare.
- Datele nu sunt clasificate apriori.
- Se incearca gasirea de conexiuni, relatii intre date, fara a avea un atribut tinta.

Semi-supervizata: Invatarea prin consolidare (Reinforcement Learning):

- Algoritmul invata prin experienta si prin feedback-ul pe care il primeste de la mediu (environment) (reward or penalty) si isi imbunatateste performanta in functie de acesta si pentru a gasi cea mai buna strategie pentru a obtine un feedback cat mai bun.

Invatarea supervizata (Supervised learning)

Tipuri <i>atribut țintă</i>	Algoritmi	Exemple
• Clasificare <i>discret</i> • Regresie <i>continuu</i>	• Metoda Bayesiană • Arbori de decizie • K-NN <i>cel mai apropiat vecin</i> • SVM (Support Vector Machine) • Retele neuronale • Regresia liniară	

Invatarea nesupervizata (Unsupervised learning)

Tipuri

- Clusterizare
- Asocieri
- Reducerea dimensionalitatii

Algoritmi

- K-means
- Clusterizare ierarhica
- Apriori

Exemple

Invatarea prin consolidare (Reinforcement learning)

- Tipuri
 - Agent, environment, reward, penalty
- Algoritmi
- Exemple
 - Game AI
 - Roboti
 - Masini autonome